

MAHAVEER APPINI

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CAREER PROFILE

Data Engineer with 4+ years of experience building high-volume ETL and real-time streaming pipelines on AWS using Python, SQL, and Kafka, with hands-on experience delivering BI dashboards on top of production data. Designs distributed data processing systems that move and transform large datasets reliably, with measurable gains in throughput, data quality, and processing efficiency across production environments. Strong in data modeling, query and pipeline optimization, and automated data validation. Holds an MS in Computer Science focused on data engineering and distributed systems, with graduate research experience at the Missouri University of Science and Technology.

EDUCATION

MS in Computer Science, Missouri University of Science and Technology

Rolla, MO | AUG 2024 - MAY 2026

B.Tech in Mechanical Engineering, National Institute of Technology

Andhra Pradesh, India | AUG 2018 - MAY 2022

TECHNICAL SKILLS

- **Languages:** C++, Python, SQL
- **Big Data & Distributed Processing:** Apache Spark (PySpark), Apache Kafka, Distributed Data Processing, Batch and Real-Time Streaming.
- **Pipelines & Orchestration:** Apache Airflow, dbt, ETL/ELT, Data Modeling, Data Transformation, Data Validation.
- **Cloud & Data Warehousing:** AWS (S3, Redshift, Glue, Athena), Snowflake, BigQuery.
- **Analytics & BI:** Power BI (dashboard development), Tableau
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, SQLite.
- **Devops & Tools:** Docker, Kubernetes, Jenkins, CI/CD, Git, Microservices.
- **Concepts:** Data Structures, Algorithms, System Design, Database Optimization, Scalable Data Systems.

WORK EXPERIENCE

Graduate Research Assistant | Missouri University of Science and Technology

Nov 2024 - May 2026

- Built and Orchestrated ELT pipelines in Apache Airflow to ingest high-frequency sensor data from **6 sources** into Snowflake, cutting the research team's data access time from two days to under an hour.
- Developed PySpark jobs to clean and transform research datasets exceeding **40 million records**, reducing processing runtime by **45%** over the prior single-machine workflow.
- Modeled raw sensor data into analysis-ready tables using dbt and built **20+** automated data-quality tests that flagged schema drift and missing values before they reached downstream analysis.
- Built GitHub Actions CI/CD pipelines to automate testing and deployment of Airflow DAGs and dbt models, running data quality tests on every pull request to catch schema drift before it reached Snowflake.
- Designed an AWS ingestion layer with S3 and Glue, exposing **200+ GB** of research data through Athena, and built **Power BI dashboards** on the curated Snowflake tables for self-service visibility into sensor trends and data quality.

Data Engineer | Capgemini

Aug 2022 - Aug 2024

- Engineered scalable ETL pipelines in Python and SQL to process millions of daily banking transactions for a major financial services client, feeding downstream reporting and risk analytics across distributed enterprise systems.
- Implemented Kafka-based streaming pipelines for reliable real-time movement of transaction data across **10+ services**, supporting fraud monitoring and asynchronous processing.
- Reduced manual data preparation by **80%** by automating ingestion and transformation pipelines for structured and unstructured financial data.
- Automated build, testing, and deployment of ETL and streaming pipelines using Jenkins CI/CD, standardizing releases and reducing manual deployment steps across compliance and operational reporting jobs.
- Optimized large-scale workflows through query tuning and structured logging, improving processing throughput by **30%**.
- Established data validation and monitoring that improved reliability and failure detection, reducing recurring data incidents by **40%** across compliance and operational reporting.

Data Engineering Intern | Capgemini

Dec 2021 - Jul 2022

- Wrote SQL queries and Python scripts to extract, clean, and prepare high-volume banking transaction data, supporting the team's ETL pipeline development.
- Created automated data-validation checks that improved accuracy and reduced downstream reporting errors.
- Streamlined recurring data-preparation tasks using Python with Pandas and NumPy, cutting manual effort for the team.

PROJECTS

Cloud-Native E-Commerce Data Platform

- Architected an end-to-end batch pipeline ingesting **10M+** e-commerce events from MongoDB and REST APIs into Amazon Redshift, modeled into a star schema, with end-to-end runtime under 15 minutes.

Real-time Log Analytics & Search Platform

- Built a GCP streaming pipeline that indexed **5M+** application logs into Elasticsearch and BigQuery, with a Redis cache cutting query latency **60%** and Kubernetes enabling horizontal scaling.